

Solutions and Decision Guide: The National Information Ecosystem

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Module purpose

Participants learn to see official statistics as a national information ecosystem rather than as the output of one national statistical office. They identify the institutions, data assets, technologies, flows, users, rules, capabilities and risks that determine whether information is fit for public decisions. They then use that map to locate realistic AI opportunities and the dependencies that must be addressed first.

Performance outcome

Given a country scenario or an approved national domain, the participant can produce a scoped ecosystem map with stable identifiers, evidence references, critical flows, user needs, quality and governance risks, coordination gaps and a justified statement of where AI may or may not add value.

Module	A1
Competencies	C1 Statistical purpose and public value; C3 stewardship; C4 quality; C6 governance; C8 communication.
Entry level	No prior AI expertise required; basic familiarity with a statistical organization or sector.
Exit level	L1 guided practitioner, with an optional L2 workplace assignment.
Structured time	6 hours, delivered as one day or two 3-hour sessions.
Practice share	Approximately 65 percent.
Primary evidence	Completed ecosystem map, flow analysis, risk statement and oral defence.
Phase 1 link	Uses the concepts and record types of Phase 1D.

Package contents

Part	Use
Use	Controlled facilitator edition. Release only after the associated exercises have been submitted or discussed.

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Fictional country case Republic of Kintaba

Kintaba is a lower-middle-income country with 24 million people. Agriculture employs a large share of the rural population and is exposed to drought, pests and price volatility. The government needs earlier district crop-production estimates for food-security planning. The following case is fictional and contains deliberate gaps and contradictions.

Actor ID	Actor	Reported role and situation
ACT-01	National Statistics Directorate	Runs the agricultural census every ten years and an annual production survey. Its frame is based on the last census.
ACT-02	Ministry agriculture statistics unit	Publishes seasonal crop estimates from extension reports and a farmer sample. Methods differ by region.
ACT-03	Extension service	Collects monthly crop-condition forms from local officers. Coverage is better near roads.
ACT-04	Land administration agency	Maintains registered parcels, but customary holdings are often absent.
ACT-05	Meteorological service	Operates 82 stations; 19 had substantial missing observations last season.
ACT-06	National food-security secretariat	Combines crop, market, nutrition and rainfall information for cabinet alerts.
ACT-07	Agricultural research institute	Runs crop-variety trials at six research stations and selected farmer fields.
ACT-08	University geospatial laboratory	Produces experimental crop maps from free satellite imagery with external project funding.
ACT-09	Market information service	Collects weekly prices by phone from selected markets.
ACT-10	Mobile network operator	Holds aggregated mobility and mobile-money indicators; no standing data-sharing agreement exists.
ACT-11	Farmer organizations	Report that official estimates arrive late and rarely include small irrigated plots or gender-disaggregated access information.
ACT-12	Government digital service	Hosts ministry systems but does not manage sector metadata or statistical methods.

Case evidence

Evidence ID	Extract
EVD-01	Statistical law assigns national coordination to ACT-01 and allows designated sector producers.
EVD-02	A memorandum requires ACT-02 to send survey microdata and metadata to ACT-01 within 90 days; last year only tables were sent.
EVD-03	ACT-02 reports 4.8 million tonnes of maize; ACT-01 reports 4.2 million for the same season. Definitions and revision status are not shown.
EVD-04	ACT-08 reports 86 percent crop-classification accuracy on 600 labelled fields in three accessible districts.
EVD-05	ACT-07 trial results show two varieties with higher mean yields under controlled water management. Adoption conditions outside trial sites are not assessed.
EVD-06	Food-security alerts are due six weeks before the annual survey estimate is released.
EVD-07	No common identifier links holdings, parcels, extension reports and survey units.
EVD-08	The geospatial project has eight months of funding remaining and one trained analyst.

Answer and decision guide

Exercise 1 reference reasoning

- ACT-01 has evidenced national coordination authority under EVD-01 and produces the annual survey. Its authority over sector release methods requires interpretation of designation arrangements, so some detail remains unknown.
- ACT-02 is a sector producer; the unfulfilled EVD-02 handoff shows a coordination and compliance problem rather than proof that its statistics are invalid.
- ACT-03 owns or generates extension reports but its proximity-to-road coverage creates selection risk.
- ACT-06 is a high-consequence user and integrator; it is not automatically the methodological authority for crop statistics.
- ACT-07 and ACT-08 are research producers whose evidence requires external-validity and sustainability assessment.
- A likely missing actor is the data-protection or security authority, treasury, regional administration, private imagery or platform provider, or a clearly named farmer subgroup. Defensible alternatives are accepted.

Exercise 2 reference assets

Asset	Fitness issue
Agricultural census and holding frame	Coverage may be obsolete between censuses.
Annual production survey	Potentially stronger statistical design but too late for early alerts.
Extension crop-condition forms	Timely but regionally inconsistent and coverage-biased.
Registered parcels	Customary holdings missing; legal parcel is not necessarily operational holding.
Weather observations	Missing station periods and spatial representativeness.
Food-security integration database or report	Depends on coherence, timing and transparent treatment of conflicting inputs.
Research trial data	Controlled or selected conditions limit population generalization.
Experimental satellite crop map	Validation narrow in geography; funding and staffing fragile.
Market price series	Selected markets and telephone collection may affect coverage and continuity.
Classifications geographies and identifiers	Critical missing integration assets under EVD-07.

Exercise 3 reference conclusion

Strong answers usually identify definition and revision incoherence between ACT-01 and ACT-02; biased and inconsistent extension reporting; missing identifiers; the six-week timing mismatch; narrow satellite validation; or fragile geospatial capability. The response must connect the chosen failure to the cabinet decision and name a responsible role. A reasonable non-AI priority is a harmonized crop and season definition, common revision status, documented early-estimate protocol, benchmark sample, flow calendar or minimum metadata exchange.

Exercise 4 decision

“Automatically produce estimates” is not ready for a pilot. A defensible decision is investigate dependencies first and narrow the use case. Candidate bounded uses include assisting analysts to document discrepancies, flagging missing metadata, summarizing reviewed field reports or testing satellite classification for selected crops and districts against an independent benchmark. Generative AI should not be the estimator or release authority. Proceed only after ownership, definitions, benchmark, access, review, security, recurrent resources and fallback are specified.